

Exercise 1

Consider an uncertain system of the form

$$x(k+1) = Ax(k) + w(k) \quad (1)$$

with $w(k) \in \mathcal{W}$. Let $\mathcal{X}$ be a nominal positive invariant set for $z(k+1) = Az(k)$ and let $\mathcal{E}$ be a robust positive invariant set for (1). Show that $\mathcal{X} \oplus \mathcal{E}$ is a robust invariant set for (1).

Exercise 2

Let $X = [a, b]$, $Y = [c, d]$ with finite $a, b, c, d \in \mathbb{R}$, $a < 0 < b$, $c < 0 < d$, and $Y \subset X$. In this case, the Minkowski sum $X \oplus Y$ and the Pontryagin set difference $X \ominus Y$ yield both again compact sets, and $X \oplus Y$ and $X \ominus Y$ both yield sets of the form $[e, f]$, $e \leq 0 \leq f$. Show that

i) $X \oplus Y = [a + c, b + d]$

ii) $X \ominus Y = [a - c, b - d]$

Exercise 3

Let $\mathcal{A}_1 := \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 2\}$, $\mathcal{A}_2 := \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$. and $\mathcal{B} := \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}$. Graphically compute

i) $\mathcal{A}_1 \ominus \mathcal{B} \oplus \mathcal{B}$

ii) $\mathcal{A}_2 \ominus \mathcal{B} \oplus \mathcal{B}$

What do you observe?